

Hormonal biochemistry and cytological changes during women menstrual cycle

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Introduction/Background

Hormonal biochemistry and cytological evaluation document the stages of the menstrual cycle and serve as an index of hypothalamic–pituitary–ovarian axis function. Hormones from the hypothalamus and pituitary gland promote follicular maturation and trigger ovulation.

The menstrual cycle is a crucial indicator of a woman's reproductive health, and any abnormalities in menstruation necessitate thorough clinical evaluation of both hormonal and structural pathology.

Study Objective

To evaluate the hormonal and cytological changes occurring during the women's menstrual cycle.

Material and Methods

Method

Vaginal cytology was used to identify the cycle phases.

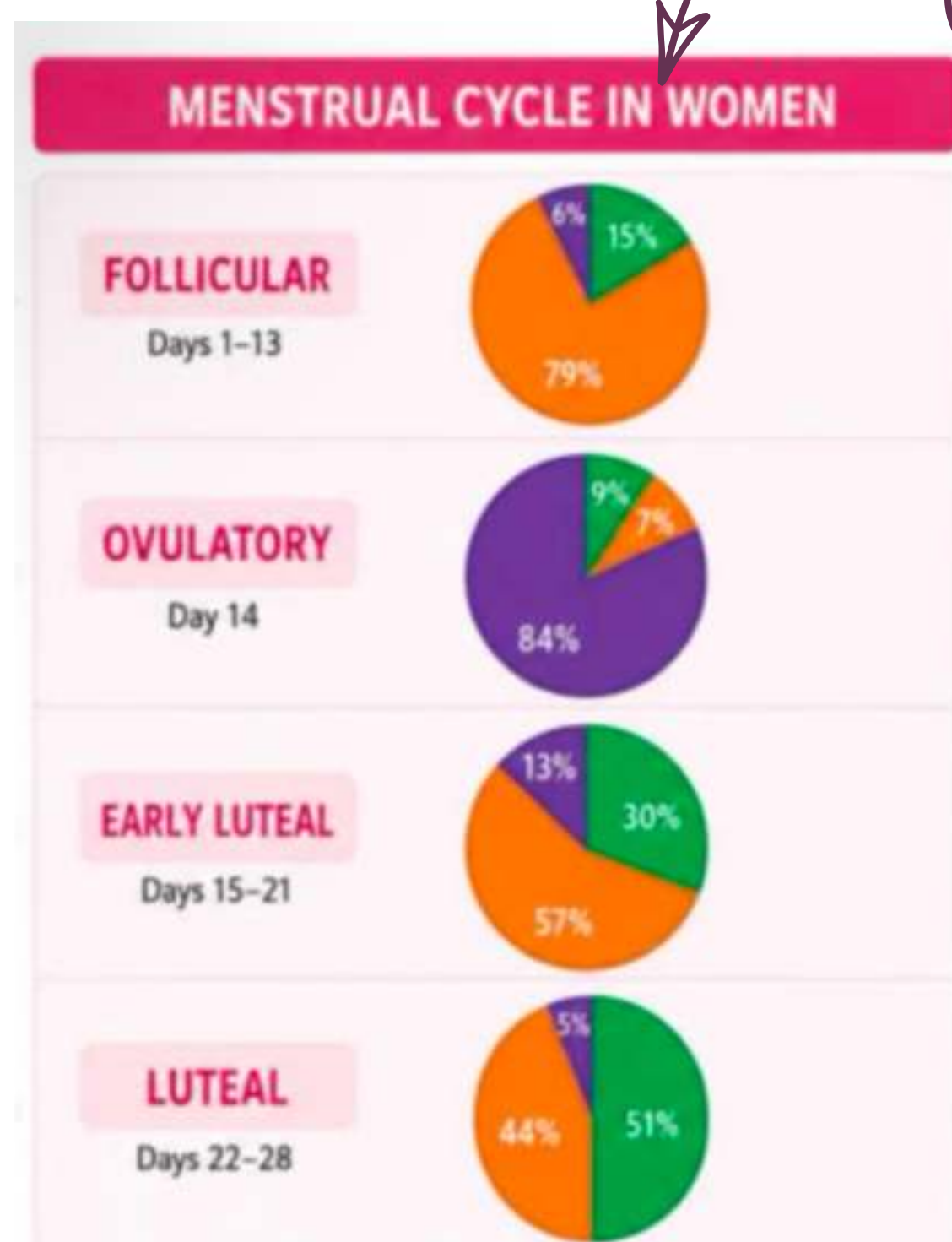
Hormones measured

Plasma levels of luteinizing hormone, follicle-stimulating hormone, estradiol and progesterone were measured across the different phases: menstruation, follicular phase, ovulation, and early luteal phase from the participants of Nafissa Hamoud Hospital (Algiers).

Procedure

The participants had an integrated two-step procedure comprising Papanicolaou smear sample collection and hormonal analysis performed during their menstrual cycle phases.

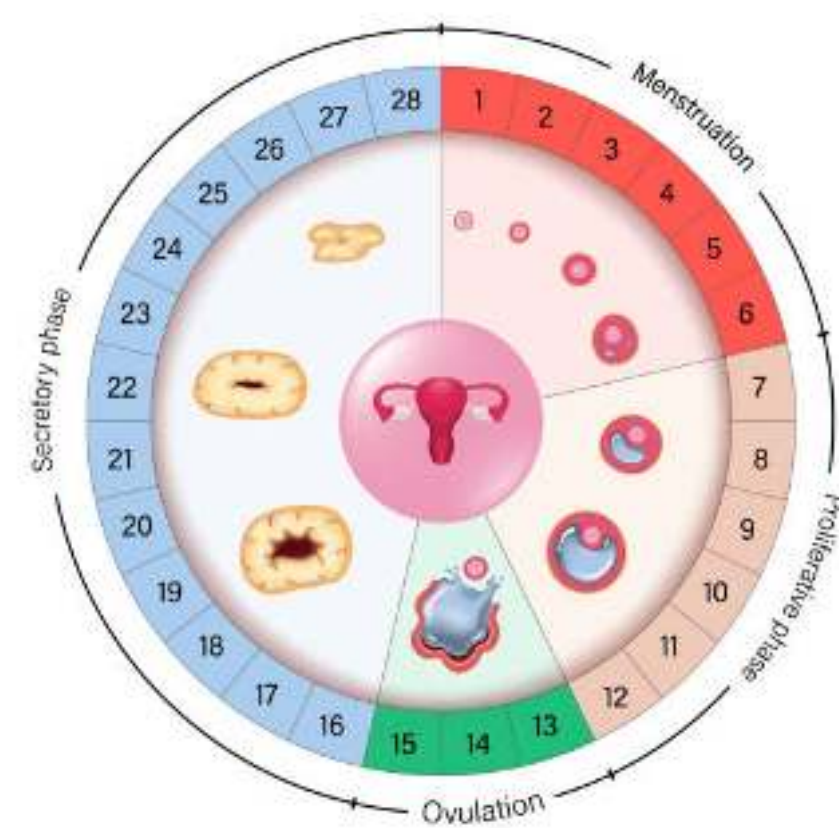
Results



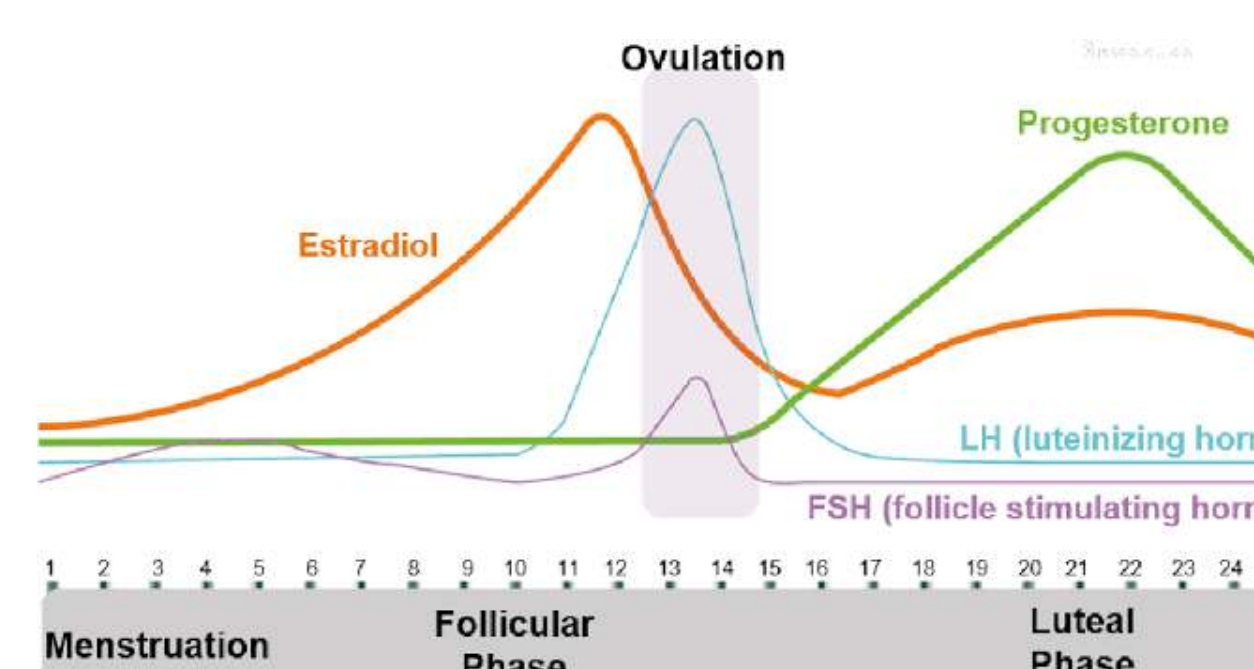
During the follicular phase, the vaginal epithelial cells are predominantly nucleated epithelial cells. However, in the ovulatory phase, epithelial cells become predominantly superficial cornified cells.

Although, throughout the luteal phase, the exfoliated cell population is dominated by leukocyte infiltration.

Women menstrual cycle

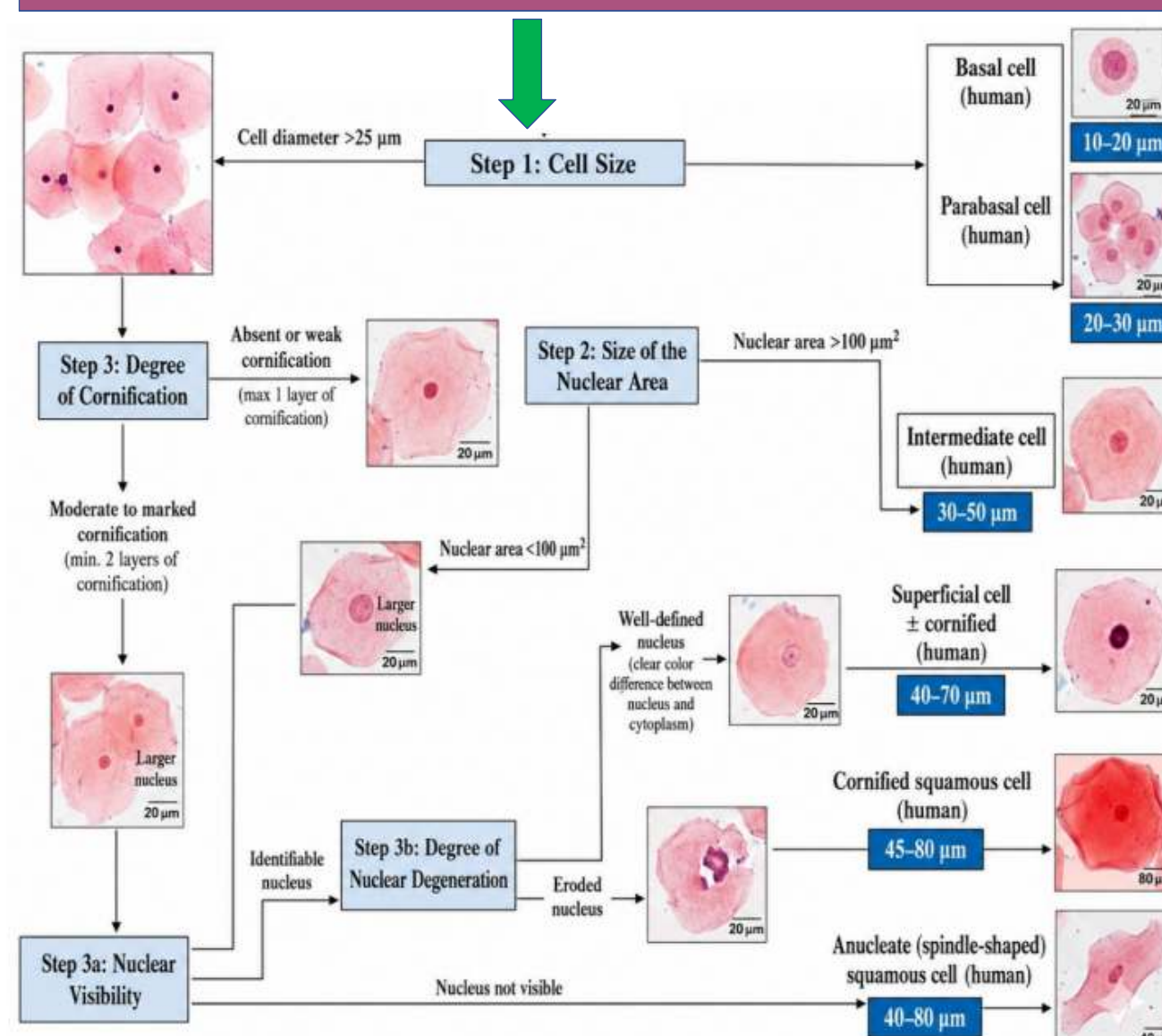


Hormonal changes of women menstrual cycle

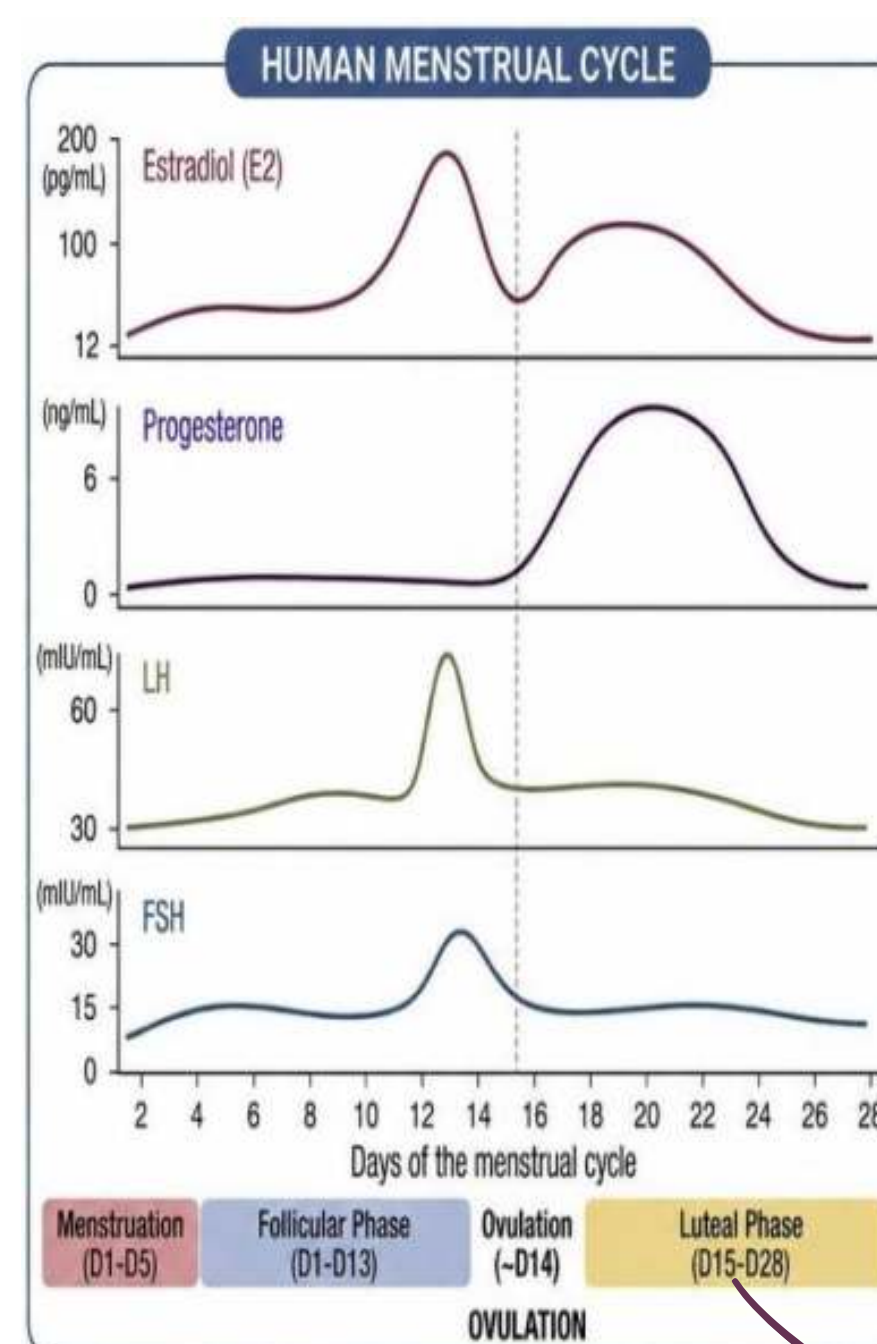
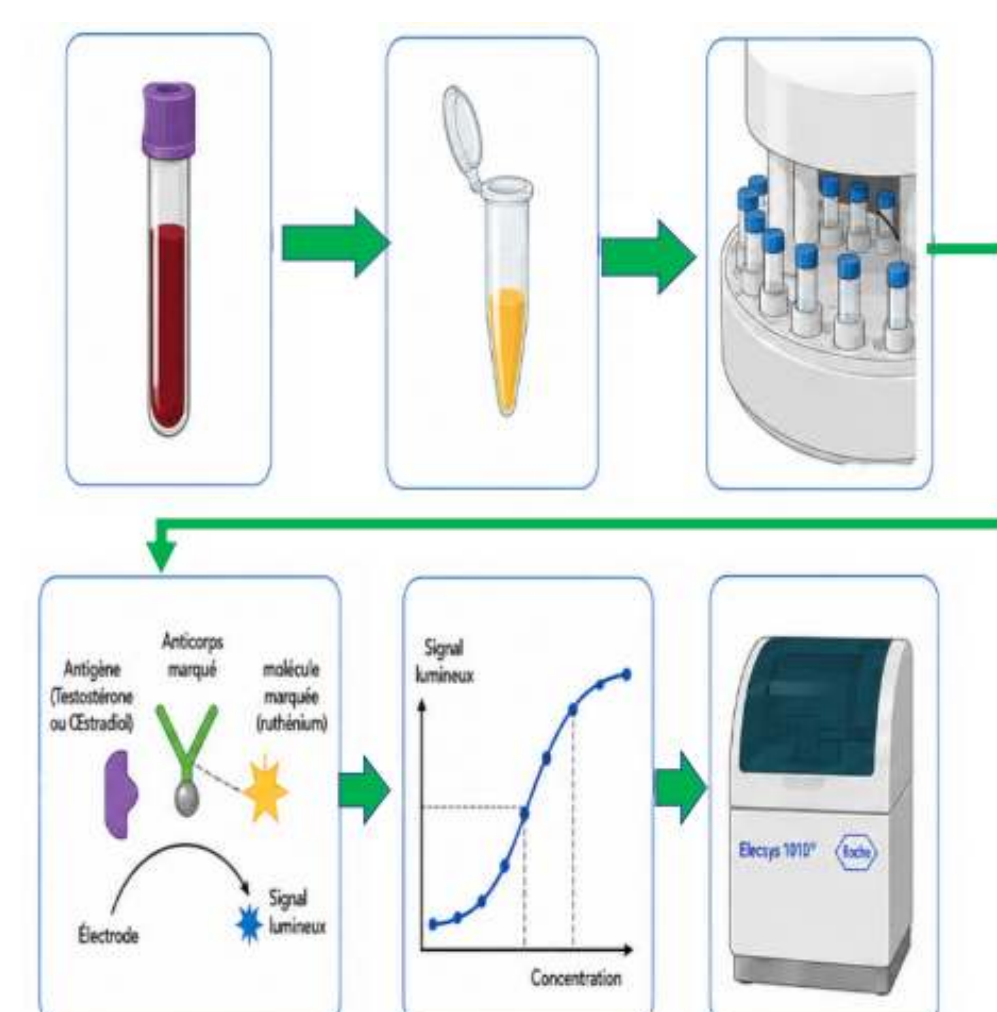


The menstrual cycle is an essential life rhythm governed by interacting levels of progesterone, estradiol, follicle-stimulating, and luteinizing hormones.

Methods for identifying Vaginal Epithelial Cells in Women



Hormonal Assay (ECLIA)



Most of the participants had a standard 28-day cycle, their menstruation begins on day 1 and lasts for approximately 5 days.

During the early follicular or menstrual phase of the cycle, levels of estradiol and progesterone are low. In the mid to late follicular phase, progesterone remains low and stable while estrogen rises to peak cycle levels. The following ovulatory phase is characterized by a surge in luteinizing hormone and a decline in estradiol.

Following ovulation, the early to mid-luteal phase is characterized by a steep rise and stabilization of progesterone along with a moderate rise and stabilization of estradiol. This is followed by the late luteal or premenstrual phase, during which estradiol and progesterone levels decline rapidly before the onset of the next menses.

Conclusively, hormonal analysis combined with cytological evaluation of the menstrual cycle provides a reliable method for clinicians to manage various gynecological issues, ultimately improving patient outcomes and quality of life.